

MATH 3203: Probability & Statistics I Syllabus

Fall 2026

Oklahoma City University, Petree College of Arts & Sciences, Computer Science

1. INSTRUCTOR INFORMATION & COURSE MATERIAL

Professor: Dr. Tashfeen, Ahmad (MSB 219B, tashfeen@okcu.edu¹).
Office Hours: Mondays, Wednesdays during 10–11 am and 5–6:30 pm or by appointment.²
Textbook: <https://www.khanacademy.org/join/5C9XBNTG>.³
Time & Location: Monday, Wednesday at 12:30–1:45 pm in MSB 201

2. TENTATIVE SCHEDULE

Week	Friday	Topic	Event	(%)
1	Aug 28	Analyzing categorical data	Homework 1	5
2	Sep 4	Displaying and comparing quantitative data	Homework 2	5
3	Sep 11	Summarizing quantitative data	Homework 3	5
4	Sep 18	Modeling data distributions	Homework 4	5
5	Sep 25	Exploring bivariate numerical data	Homework 5	5
6	Oct 2	Study design	Homework 6	5
7	Oct 9	Probability	Homework 7	5
8	Oct 16	Everything Covered	Midterm	15
9	Oct 23	Counting, permutations, and combinations	Homework 8, Fall Break	5
10	Oct 30	Random variables	Homework 9	5
11	Nov 6	Sampling distributions	Homework 10	5
12	Nov 13	Confidence intervals	Homework 11	5
13	Nov 20	Significance tests (hypothesis testing)	Homework 12	5
14	Nov 27		Homework 13, Thanksgiving	5
15	Dec 4	Inference for categorical data (chi-square tests)	Homework 14	5
16	Dec 11		Reading Week	
17	Dec 18	Everything Covered	Final	15

TABLE 1 — This may change depending upon class progress.

3. GRADING CRITERIA

When accepted, work that is $n < 3$ days late will receive no more than $100(1 - 0.1n)\%$ of credit (10% deduction per day). For certain work, late submissions will not be accepted. Any submission that is not easy to read will receive a zero.

Total $t\%$	Grade	Total $t\%$	Grade	Total $t\%$	Grade
		$t \geq 93$	A	$93 > t \geq 90$	A ⁻
$90 > t \geq 85$	B ⁺	$85 > t \geq 82$	B	$82 > t \geq 80$	B ⁻
$80 > t \geq 75$	C ⁺	$75 > t \geq 72$	C	$72 > t \geq 70$	C ⁻
$70 > t \geq 65$	D ⁺	$65 > t \geq 62$	D	$62 > t \geq 60$	D ⁻
$60 > t$	F				

TABLE 2 — A grade of Incomplete (“I”) will only be assigned in case of documented extenuating circumstances. The “I” will be removed in accordance with university policy stated in the online undergraduate catalogue. Please also see subsection 4.4.

¹Prefix [MATH3203] followed by a space to all emails sent to the instructor.

²Zoom for a virtual meeting.

³Use your school’s email address and name as it appears at the online classroom.

4. POLICIES & RESOURCES

4.1. General Academic Guidelines. For an up-to-date version of the guidelines please visit the university [sharepoint](#), see the [OCU course schedule website](#) for the finals week schedule and visit the [online classroom](#).

4.2. Course Description. The class guides students from basic data description to formal statistical inference. Core topics include categorical data analysis, quantitative data visualisation, comparison, and summarisation; probability foundations (basic rules, counting, combinatorics, random variables, and sampling distributions); and inferential techniques such as confidence intervals, hypothesis testing, chi-square methods, and simple linear regression. The course culminates in the design and evaluation of empirical studies.

4.3. Course Learning Outcomes.

- Describe and summarize categorical and quantitative data using appropriate graphical and numerical techniques.
- Apply probability rules, combinatorial methods, and sampling distribution theory to model random phenomena.
- Conduct inferential procedures—including confidence intervals, hypothesis tests, and chi-square analyses—to draw conclusions from data.
- Develop and assess bivariate regression models and study designs for rigorous statistical investigation.

4.4. Academic Integrity. The cheating rule for this class is simple: *don't turn in anything you don't understand*. I encourage you to get help and exhaust your resources. However, if you turn in something (or answer a question with something) you do not understand and cannot explain when I ask you to do so, you will receive an *automatic F in the course*, and disciplinary action will be taken. If I ask you to demonstrate your work, you will need to do so during my office hours or by appointment. If you fail to demonstrate it, you will receive a zero on the assignment. For more, read the *Academic Honesty* section of the [courses' catalogue](#).

4.4.1. Artificial Intelligence (AI). Help obtained from an AI, e. g., a large language model chat agent, should be documented and turned in with the work. You may use AI to understand a general concept, but it should *never* write or solve anything for you. A student who abuses AI will forfeit the right to use it for the remainder of the course.

4.5. Religious Accommodation. Oklahoma City University seeks to be supportive of religious observance among the members of our diverse campus community and to be as accommodating as possible. Students should discuss with their instructor at the beginning of the semester forms of religious observance (dress, fasting, specific prayer times) that may affect their full participation in the course. Students should also compare the class schedule to their own religious calendar to determine if there will be any class days in which the student expects to be absent due to the observance of a religious holiday. *Students must notify the instructor, in writing, of the expected absence within the first two weeks of the semester.* The instructor will then work with the student to develop

a plan to reschedule any exams, assignments, or course activities for that day. The instructor, at his/her own discretion, will make reasonable accommodations wherever possible. Students should recognize that there may be some course aspects that cannot be rescheduled or accommodated, and it will therefore rest upon the student to determine whether they wish to remain enrolled in the course or have their grade potentially affected.

4.6. Mission Statement. The Petree College of Arts and Sciences provides a supportive, student-centered learning environment. Through an interdisciplinary curriculum that promotes active learning and individualized instruction, students are encouraged to think broadly, to find their passion, to collaborate with others, and to connect with the broader community.

4.7. Grook.

Put up in a place
where it's easy to see
the cryptic admonishment
T. T. T.

When you feel how depressingly
slowly you climb,
it's well to remember that

Things Take Time
—Piet Hein (1970)